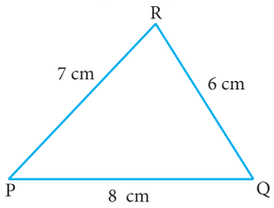
Example 4.12 Construct the centroid of ΔPQR whose sides are $PQ = 8$cm; $QR = 6$cm; $RP = 7$cm.

Rough Diagram



Solution

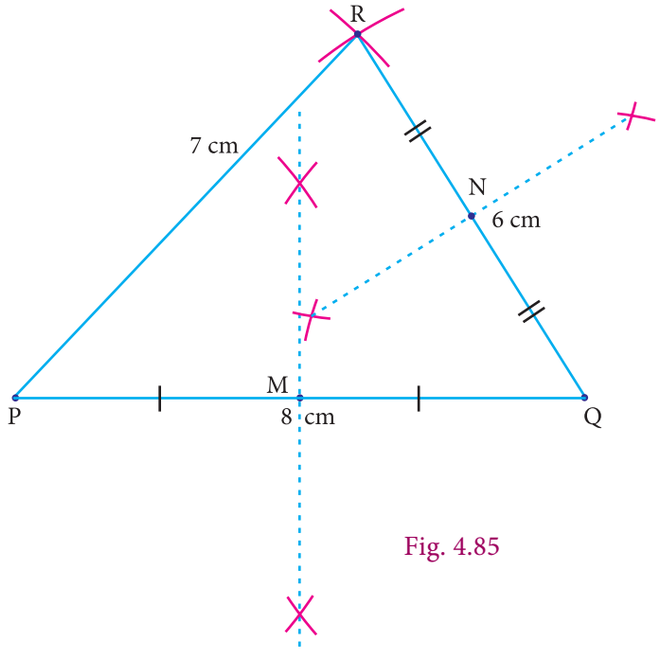


Fig. 4.85

Step 1 : Draw ΔPQR using the given measurements PQ = 8cm QR = 6cm and RP = 7cm and construct the perpendicular bisector of any two sides (PQ and QR) to find the mid-points M and N of PQ and QR respectively.

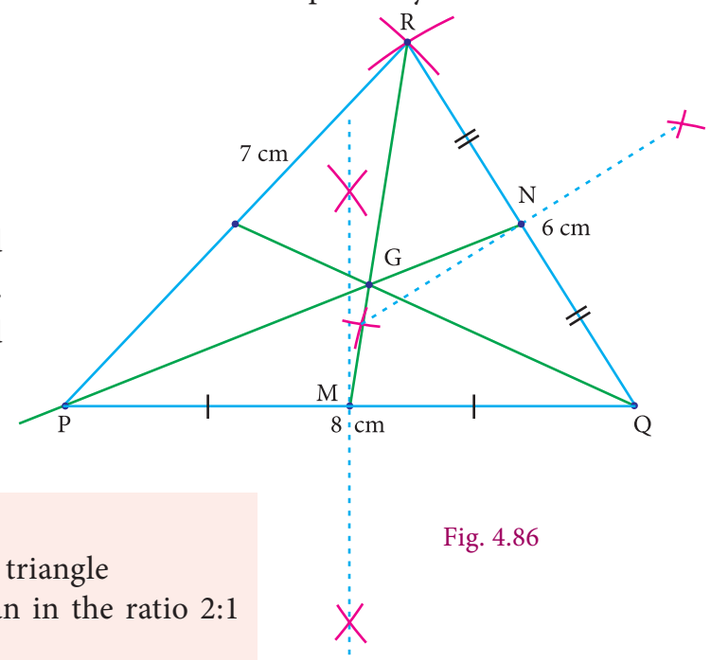


Fig. 4.86

Step 2 : Draw the medians PN and RM and let them meet at G. The point G is the centroid of the given ΔPQR.

Note

- Three medians can be drawn in a triangle
- The centroid divides each median in the ratio 2:1 from the vertex.
- The centroid of any triangle always lie inside the triangle.
- Centroid is often described as the triangle's centre of gravity (where the triangle balances evenly) and also as the barycentre.

4.7.2 Construction of Orthocentre of a Triangle

Orthocentre

The orthocentre is the point of concurrency of the altitudes of a triangle. Usually it is denoted by H.

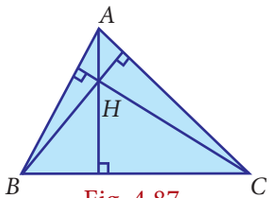


Fig. 4.87

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